A REVIEW OF HISTORICAL AND CURRENT CODING STANDARDS FOR OPTICAL COMMUNICATION

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ABSTRACT. A survey of some significant forward error correcting codes used in optical communication is presented. The report presents advancements in coding as a consequence of the "scaling problem": increasing the scale and throughput of optical networks induces the continuing need for low-latency, power-efficient, and highly accurate error correction (high coding gain at extremely low bit error rates). Academic literature and industrial standards are reviewed to understand major FEC schemes starting with the (255,239) Reed-Solomon code recommended by the ITU-T in G.975 in 1996. Turbo Product Codes and Zipper codes are also explored toward concluding with two schemes widely implemented in optical networks, open- and concatenated-FEC. The report serves to concisely review some important standardized codes in the history of FEC in optical communications.

CONTENTS

1. INTRODUCTION

Fiber optic has been an integral part of global telecommunication backbones since the 1990s, and in the United States particularly since the 1980s. Forward error correction (FEC) in local or metro area optical networks (on the order of hundreds of Mbits/second) was not a major concern since throughput was not high enough to induce poor signal-to-noise ratios (SNR). By the 1990s however, optical networks were operating at up to 2.5 Gbps, a rate at which FEC could no longer be ignored. The first published recommendation for the use of FEC in optical networks appears in the 1996 ITU-T (International Telecomunnications Union - Telecommunication Standardization Sector) G.975 Recommendation for submarine systems [12]. Later, the 2001 ITU-T G.709 Recommendation for interfaces for optical transport networks (OTN) [13] also standardized FEC in optical networking. The schemes presented in these early standards demonstrated the potential to make clever use of FEC to scale throughput while maintaining a low bit error rate (BER). Indeed, FEC capabilites have only increased with ever-increasing global traffic on optical channels of all scales.

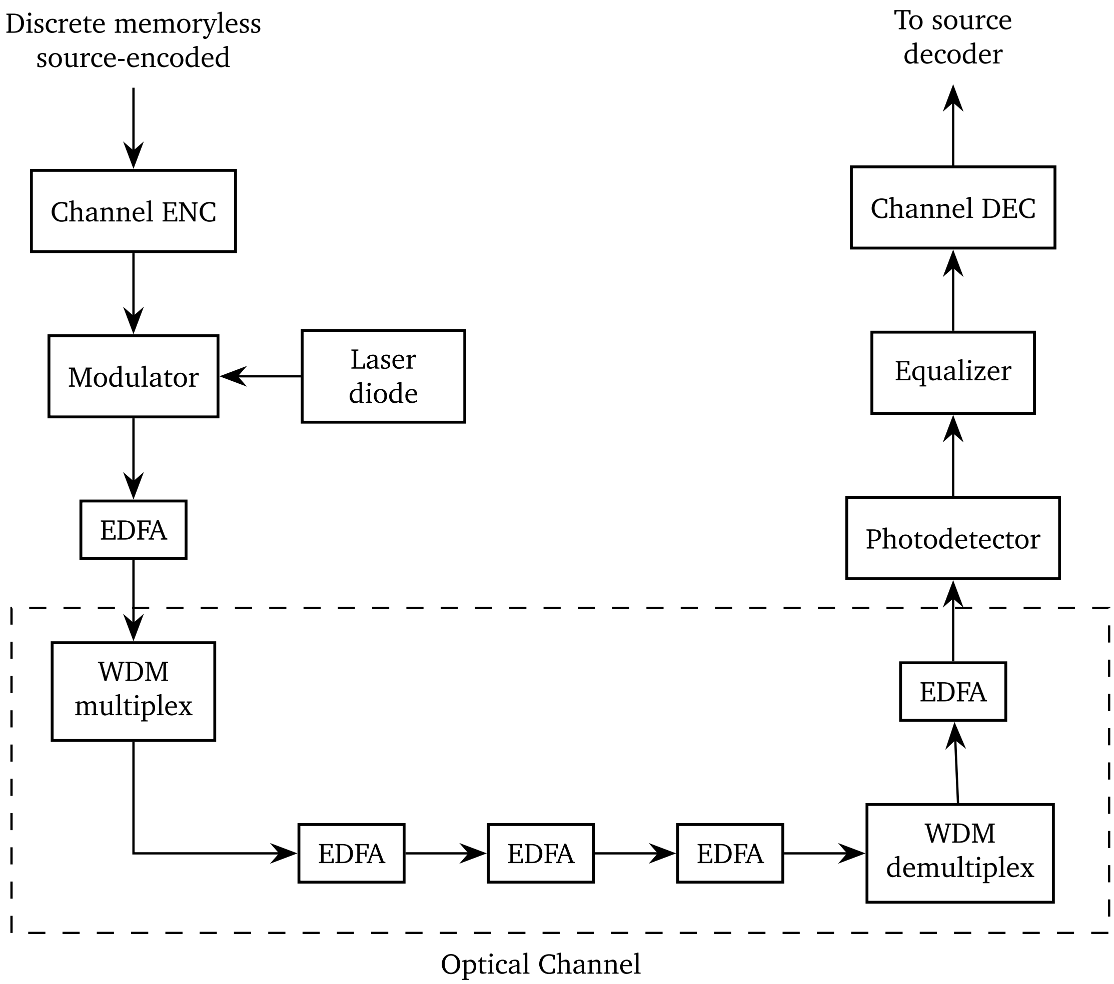

FIGURE 1. Point-to-point digital communications system after [4]

Figure 1 depicts a standard digital optical communication system as illustrated in [4]. Clearly, channel coding is only part of the overall system, but an important one. Modulators transform the coded data into light, multiplexers divide independent data streams into several wavelengths that will not interfere with one another. EDFA (erbium-doped fiber amplifier) are optical amplifiers that allow the signals to be carried and repeated over long distances.

Generally, advances in FEC for optical channels are driven by this scaling problem: as channel throughput increases, signals suffer from poorer SNRs due to harsher channel conditions. To ensure messages are reliably received after transmission through these harsher channels, more robust FEC algorithms are required. Effective FEC algorithms generally consume more power due to a higher decoding complexity, which is not ideal for large-scale and long-haul communication. There is thus a continuing need to develop effective FEC that efficiently (with respect to power consumption and latency) decodes noisy signals with exceptional accuracy, usually at BER on the order of 10^{-12} to 10^{-15}.

This report explores such advances in FEC beginning with the aforementioned ITU-T G.975 recommendation of a forward error correcting Reed-Solomon code. Further codes, standards, and frameworks will be examined toward the presentation of two widely-implemented modern schemes that are still used today: open-FEC (o-FEC) and concatenated-FEC (c-FEC). Before turning to Reed-Solomon codes, common metrics for comparing code performance are presented, namely coding gain and overhead.

Definition 1.1. Coding gain (sometimes denoted NCG for "net coding gain" and expressed in dB) is the difference between the signal to noise ratio per information bit required to achieve a desired bit error ratio (BER) with FEC and without FEC. This is the coding *gain*, since there is an improvement upon receiving the code with all unwanted noise.

Example 1.2. Suppose that over some channel in a noisy environment, we want to examine the coding gain of some FEC scheme at the BER level of 10^{-9}, meaning upon receipt of transmission at the receiver, only $1/10^9$ bits are erroneous. Suppose that without any FEC (transmitting the signal directly over the channel) an SNR of 10.0 dB is required to achieve the 10^{-9} BER. This is a very high signal-to-noise ratio, meaning more signal needs to be sent to obtain a reliable transmission. Suppose that after encoding the data with some error-correcting code prior to transmission, we require an SNR of only 3.0 dB to achieve that same 10^{-9} BER. The intuition is that there is less signal power needed per noise power to obtain the same result. The gain is computed as

$$G = \left(\frac{E}{N}\right)_0 - \left(\frac{E}{N}\right)_1 = 10.0 - 3.0 = 7.0 \text{ dB},$$

where the subscripts 0 and 1 denote the null transmission (without FEC) and the alternative transmission (with FEC).

Signal to noise ratio E/N and rate k/n are familiar measures in signal processing and information theory. For practical consumer and commercial applications, the rate of data transmission is typically described in terms of bits per second: Kbps (10^3 bits/s), Mbps (10^6 bits/s), or Gbps (10^9 bits/s). Similarly, OTN throughput rates are described by 100G, 200G, 400G, etc., where G dentoes gigabits per second; these are the rates at which source-encoded message data are fed to a channel encoder. Increases in this channel rate are precisely what induces the scaling problem.

Definition 1.3. Overhead is the ratio of redundant bits added by an error correcting code to message bits. The overhead for a (n, k) code can be expressed as

$$\frac{n-k}{k}.$$

Most often, overhead is expressed as a percentage: an overhead of 5% means that one parity bit is added for every 20 message bits.

2. Reed-Solomon Codes

The 1996 ITU-T G.975 recommendation [12] presented the first standardized FEC scheme for optical communication. The FEC proposed is a RS code of length 255 and dimension 239 having rate $239/255$. RS codes were well-studied had applications in small electronics and in extreme conditions like deep space communication. RS codes have a nice algebraic description [10] [1] and thus make an excellent baseline FEC scheme due to simple encoding, low-complexity decoding, and a good capability to correct burst

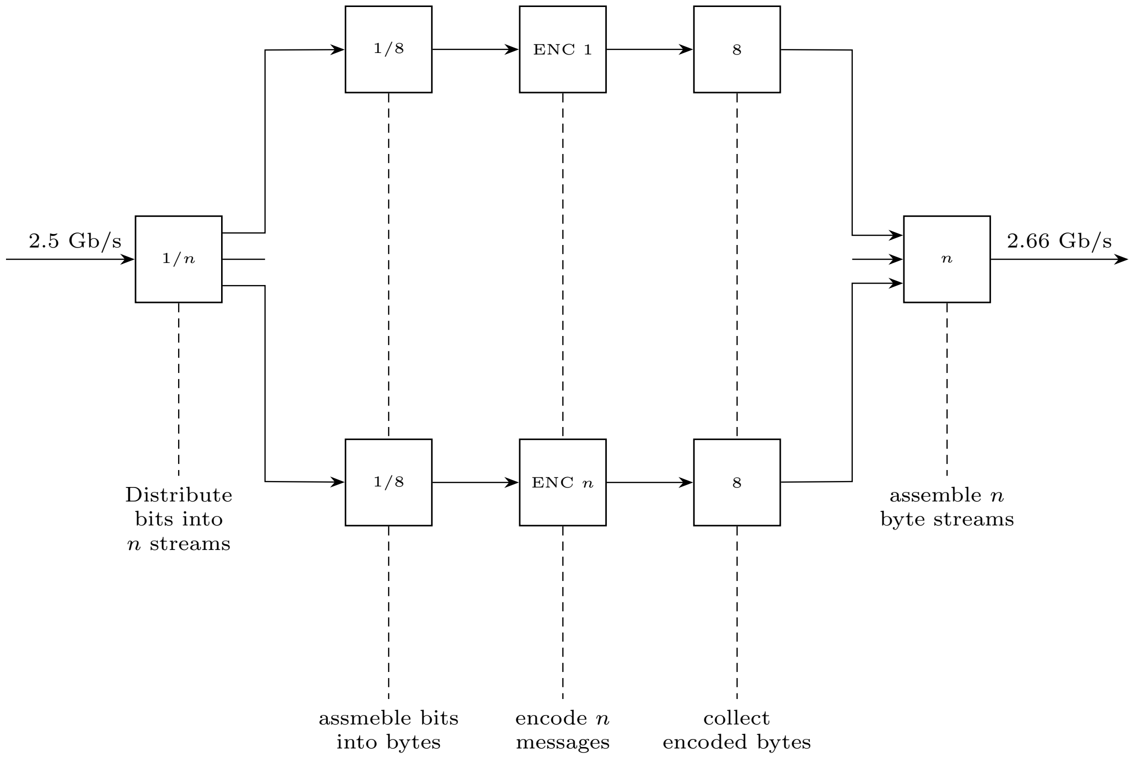

FIGURE 2. Encoder using n interleaved RS(255,239) encoders after [12].

errors. Burst error correction is of particular concern in long-haul optical networks due to a tendency to amplify errors over long distances. The code is defined over the Galois Field $\mathbb{F}_{256}$ by the generator polynomial on the conjugates of the primitive element α:

$$f(x) = \prod_{i=0}^{15}(x - \alpha^i)$$

This RS code is a good choice for OTN encoding and decoding due to its ability to correct up to 8 error symbols ($8 \cdot 8 = 64$ bits), since

$$t = \left\lfloor \frac{n-k}{2} \right\rfloor = \left\lfloor \frac{255-239}{2} \right\rfloor = \frac{16}{2} = 8.$$

While the recommendation does not prescribe a decoder, it does cite the low decoding complexity as a reason for recommending RS(255,239). It is well known that any practical decoder including the celebrated Berlekamp-Massey, Berlekamp-Welch, or Euclidean decoding algorithms are perfectly acceptable algorithms with complexities on the order of $O(n \log n)$ to $O(n^2)$.

The recommendation also cites the burst error correction capacity as a reason for recommendation; an encoder is presented with 16 interleaved RS(255,239) codes. Figure 2 depicts the interleaving process of n RS codes after [12]. A message stream of bits is distributed among n streams, each of which is separated into 8 bit streams; this is a result of the RS(255,239) code working with elements of $\mathbb{F}_{256}$ which can be expressed as sequences of 8 bits. The output of such an interleaved encoder is called an FEC frame.

Since there are 8 bits for each of n codes, each frame has a length of $255 \cdot 8n = 2040n$ bits. The decoder proposed in the recommendation is strictly symmetric to the encoder in Figure 2 toward an effort of simple and straightforward decoding.

The G.975 recommendation was succeeded by the revision G.975.1 in 2004, but the RS(255,239) was not made obsolete. The updated recommendation presented various "super-FEC" codes and made use of the revolutionary *Turbo decoding*.

3. Turbo Codes

In 1993, Claude Berrou introduced a new class of codes called Turbo codes [2]. Berrou presented the classical implementation of a parallel concatenated convolutional code with recursive systematic encoders, depicted in Figure 3.

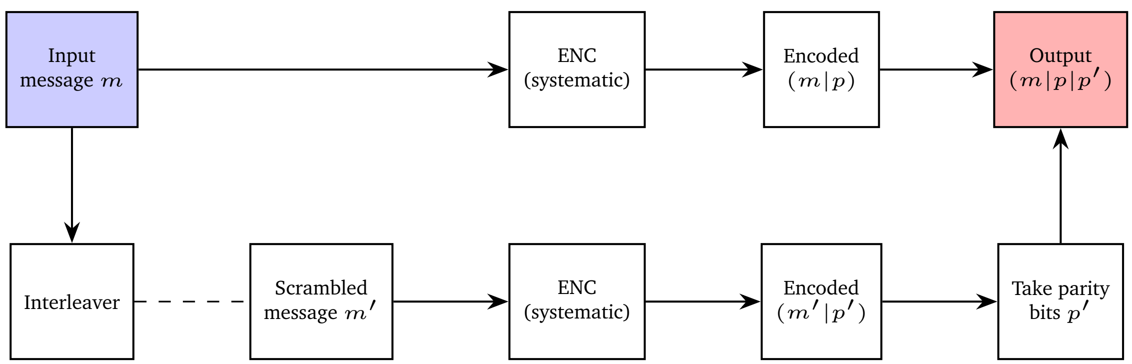

FIGURE 3. Classical Turbo PCC code

The contribution of Berrou's Turbo codes toward optical communication was not the classical convolutional code originally presented, since optical networks demand extremely high throughput rates with low BER which PCC Turbo codes do not scale well with. The real revolution in Berrou's work was in decoding: turbo decoding brought coding gain very close to the Shannon limit, even in practice. By using a maximum likelihood decoding algorithm with some fixed number of iterations (usually 3-5 iterations), FEC is improved significantly. Turbo decoding was adapted to simpler product codes for use in optical networks, emerging as a successor to RS codes. A Turbo Product Code (TPC) uses Turbo decoding on a simple product code of linear constituents. An early example of a TPC appearing in standards literature was in a 2000 submission to IEEE 802.16 for wireless metro area networks [16]. Several block turbo codes of different dimension were submitted during a call for contributions, citing a higher BER performance, higher code rates for increased throughput, and a universal encode/decode design at no added complexity. Though this was proposed for wireless and not optical networks, it demonstrates the direction that FEC was moving in at the turn of the century. By 2004, the ITU-T superseded its original G.975 recommendation, replacing it with the G.975.1 recommendation [11]. In this document, "super-FEC" schemes are presented as having better FEC than the original RS(255,239) codes; one code presented is a concatenated code using, simply, an outer RS(1901,1855) and an inner TPC. The inner code is a product of (512,502) and (510,500) Hamming codes that uses a fixed number of soft decode iterations at the inner decoder.

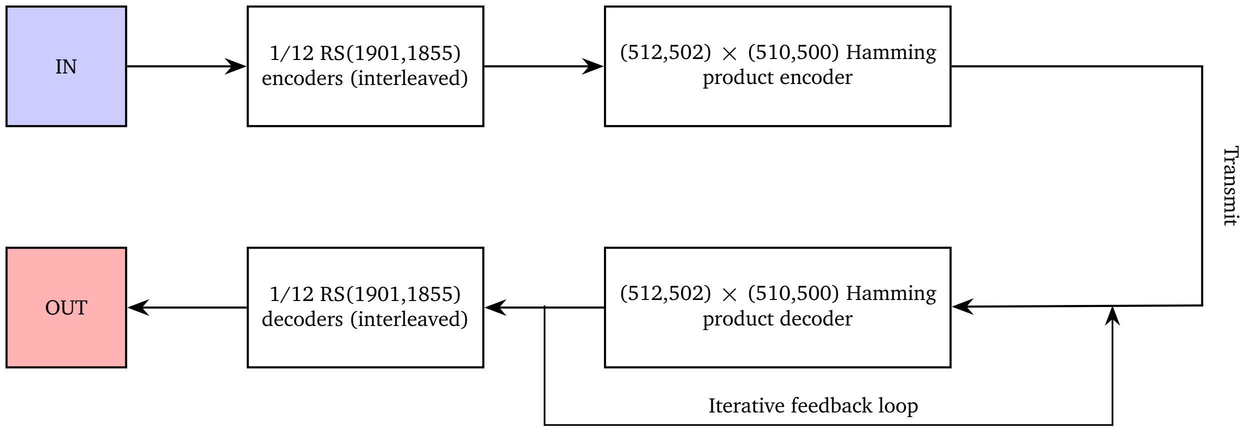

FIGURE 4. Super-FEC scheme in ITU-T G.975.1 [11]

4. ZIPPER CODES

A zipper code is one code partitioned into two sets (zipping pair) along with an interleaver. Given a code C of length n and dimension k, a sequence of codewords $c_0, c_1, c_2, \dots, c_m \subset C$ is called a buffer. Then for every codeword in the buffer, we can consider all codeword entries up to an arbitrary index $j \leq k$. The union of these codeword slices is called the *virtual set* of the constituent. This notion admits a complementary notion called the *real set*, which is simply the remaining codeword symbols after the arbitrary index j.

Example 4.1. Let $C = \{000, 011, 101, 110\}$ be the binary $(3, 2)$ code. A possible virtual set is
$$A = \{00, 01, 1, 11\}$$
with complementary real set
$$B = \{0, 1, 01, 0\}.$$
The indicies that determine where the split happens are $\{2, 2, 1, 2\}$. This partition can be visualized as in [15] via Figure 5

Notice that the redundancy bit on each codeword is immediately in the real set. This is always the case, and is a consequence of the restriction of $k < n$ on the splitting positions.

The next core component of a zipper code is an interleaver map which associates to each codeword bit in the virtual set a codeword bit in the real set. Though a bijective interleaver is not immediate, we will only consider bijections in this report. The interleaver does not take bits alternating from the virtual and real sets in the naive sense, rather it serves to complete the first part of each codeword depending on the values of the second part. Each codeword is encoded by filling the latter part (corresponding to the virtual set) with a message to be transmitted. Then the interleaver fills in the remaining first part of the codeword by drawing a symbol from the image of the interleaver, which was predetermined when defining the map. Lastly, add on parity symbols to the end of the codeword to yield a full codeword that can be transmitted and decoded upon receipt.

Example 4.2. Suppose we wish to encode into a $(7, 4)$ zipper code at some codeword c_i with $A_i = \{1, 2\}$ and $B_i = \{3, 4\}$. Say we receieve bits $(0, 1)$ from a stream of message data

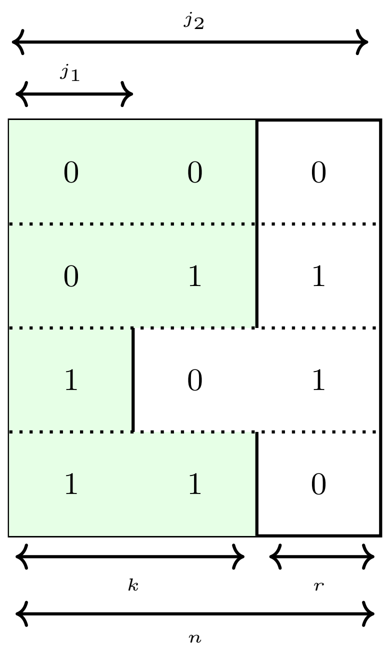

FIGURE 5. Visualization of zippering in a block code after [15]

and have a predetermined interleaving rule for this ith codeword given by

$$\begin{aligned}\phi\colon 0 &\mapsto 3\\ 1 &\mapsto 2\end{aligned}$$

The encoding proceeds as follows:

(1) Insert message bits into the real positions indexed by B_i.

$$c_i = \_\_01\_\_\_$$

(2) Insert interleaved mappings of real bits in virtual positions A_i.

$$c_i = 1001\_\_\_$$

(3) According to some parity rule, fill parity positions $\{4, 5, 6\}$.

$$\begin{aligned}p_1 &= c_1 + c_2 + c_3 = 1 + 0 + 0 = 1\\ p_2 &= c_2 + c_3 + c_4 = 0 + 0 + 1 = 1\\ p_3 &= c_1 + c_2 + c_4 = 1 + 0 + 1 = 0\end{aligned}$$

So that the final encoded word is $c_i = 1001110$.

5. MODERN FEC

The 2010s brought the era of "terabit" ethernet (100G and above). In 2012, staircase codes were proposed in [14], which provided a low-latency scheme specifically for 100G OTN. In the decade following, coding schemes of gain 11 dB or higher were designed and implemeneted, namely concatenated FEC (c-FEC) and open FEC (o-FEC). These codes are still widely implemented in coding for optical communication today, along with LDPC codes.

5.1. **c-FEC.** Concatenated codes have been widely studied and implemented in different ways. Concatenated codes use an *outer* and an *inner* code, effectively protecting the message bits twice: in encoding, the message bits are wrapped by the outer encoder, and the resulting outer codeword (including parity bits) are wrapped by the inner code. Counterintuitively, the inner code can be imagined as an outer shell that protects the original encoding during tranmission, while the outer code can be thought of as an inner shell that adds further protection in case of severe noise. Section 2 discussed the super-FEC scheme from 2004 which was a concatenated code of outer RS(1901,1855) and inner Hamming product of (512,502) $\times$ (510,500).

This section presents the modern c-FEC code as in the 2020 OIF Implmentation Agreement for 400ZR interfaces [7]. This c-FEC uses an outer (512,510) Staircase code with BCG(255,239) constituents, which was introduced in 2012 in [14], specifically for 100G OTN applications. Staircase codes have excellent efficiency and low latency; pairing them with a simple inner Hamming (128,119) code maintains exceptional decoding speed. This FEC scheme is depicted in Figure 6, and notes the soft-decision decoding trait of the inner Hamming code. while not included in all applications, three decoding iterations contribute to the net coding gain of the c-FEC code. Specifically, at the 10^{-15} BER level soft-decision with no feedback iterations results in a NCG of appproximately 10.8 dB; including three feedback iterations increases this to 11.1 dB, a significant gain.

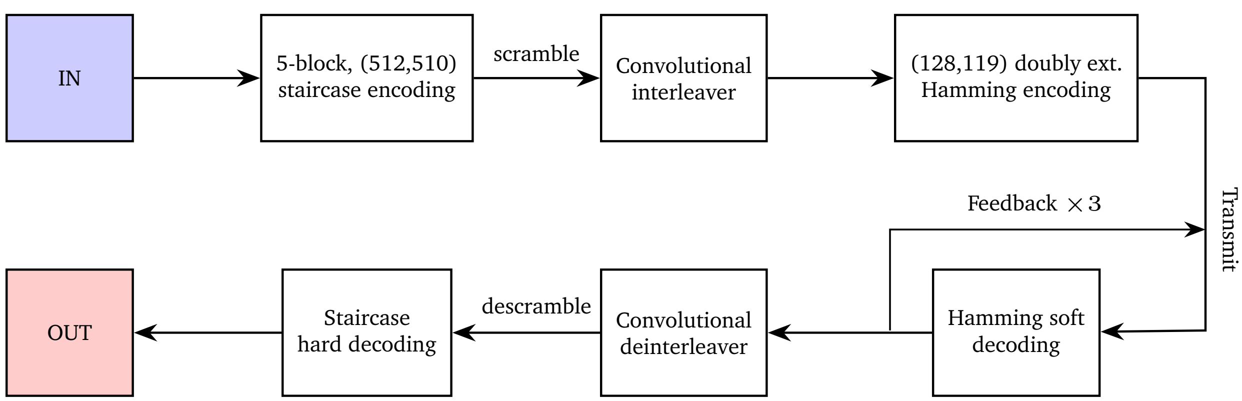

FIGURE 6. Concatenated FEC (CFEC+) scheme in OIF 400ZR [7]

The ultra low latency of this c-FEC code lends itself well to short range applications, noted in the OIF 400ZR as being rated for less than 120 kilometer distances. c-FEC are used extensively in local area 5G applications and serve as the short-range "sprinter" of modern FEC.

5.2. **o-FEC.** Open FEC codes make use of several of the topics presented in this report so far: o-FEC have a staircase-like semi-infinite matrix structure, can be expressed in a zipper-like representation, and are a form of Turbo Block Code. In contrast to c-FEC, o-FEC specifically does not fully comply with the original code rate of 239/255 as specified in the G.975 and G.709 ITU recommendations, to offer more flexibility and lift constraints that suppress high coding gains. Indeed, o-FEC uses a (256,239) extension of a (255,239) BCH as a constituent code.

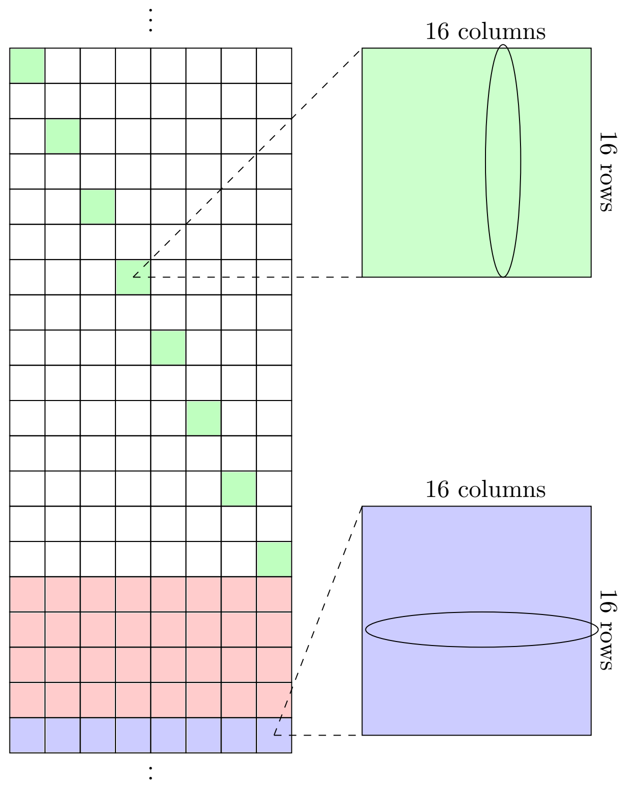

FIGURE 7. Open FEC infinite matrix encoding

Open FEC codes are specified in the Open ROADM MSA for 100G to 400G networks over coherent optical links [9]. o-FEC are characterized by block-based encoding and soft-decision decoder with feedback iterations (Turbo decoding algorithms are recommended but not specified in the document). Constituent component o-FEC codewords are encoded in an iterative staircase-like spatially-coupled way, and in a zipper-like way that clearly partitions each codeword into a front and back part. The following example is based upon an example appearing in [9] and illustrates the encoding process.

Example 5.1. Constituent component codewords are identified by an ordered pair (R,C), where R corresponds to one of an infinite number of rows, and C corresponds to one of 8 columns. The ordered pair identifies block matrix entries in the semi-infinite matrix depicted in Figure 7, where each block is a 16×16 matrix of bits.

This example will demonstrate identifying the constituent component codeword $(R, C) = (20, 10)$. First, identify the $R = 20$th row of the semi-infinite matrix; this row is identified by the blue row in Figure 7. Next move up by a buffer of an even number of rows, called the guard; this guard is identified by the four red rows in Figure 7. In the first row immediately following this guard, take the rightmost block matrix. Ascend the semi-infinite matrix by moving back by one column and up by two rows until the first column is reached. The block matrices identified with this step are represented by the green block matrices in Figure 7.

After identifying each of the blue and green matrices, the codeword bits can be read directly. The front of the codeword (positions 1 through 128) are the bits appearing in the

10th column of each green block matrix. The remaining back of the codeword (positions 129 through 256) are the bits appearing in the 10 row of each blue block matrix. This creates the final BCH constituent component codeword of length 256 as desired.

The final codeword is identified in a zipper-like structure, where every codeword is split evenly into a first half and second half, called the front and back, respectively. For every codeword, bits in the front appear exactly once in the back of another codeword, and bits in the back appear exactly once in the front of another codeword. It is clear that each bit in the code is thus protected twice. Sliding the template in Figure 7 along the semi-infinite matrix provides the codewords for the o-FEC code. Each window of this form contains 16 codewords, each corresponding to one of 16 rows/columns in each block.

In o-FEC, two parallel encoders are used and both use the method described in the preceding example. Exactly half of the bit stream enters each of the encoders, determined by the parity (i.e. 1- or 0-based, not in the sense of parity bits added by the encoder). Decoding is also done twice due to the parallel encoding described, and the doubly-protected nature of each bit. The decoding begins by processing the back of a given codeword (the last 128 bits including all 17 parity bits). After one full decoding iteration when all bits are decoded, a second iteration is performed in an identical way, except considering the full codeword including both front and back parts [6]. Chase provides a standard decoding algorithm for block codes [3], and this applies to o-FEC here.

5.3. **LDPC.** Low-density parity-check codes are characterized by their sparse parity check matrices, often with a proportion of zeros greater than 99%, resulting in fewer floating point operations and ultimately lower computational complexity than other common FEC schemes. Gallager originally developed the theory of LDPC in the 1960s [8] but only entered practical and research prominence in the 1990s after Berrou's Turbo codes were introduced [2].

In the ITU-T G.975.1 Recommendation that proposes several super-FEC schemes following the original RS(255,239) code, a LDPC code is proposed [11]. From 2004, this is a very early appearance of LDPC codes in optical networks. The code presented is a systematic binary LDPC(32,640, 30,592) and is specifically designed to be identical in rate to the original interleaving of 16 RS(255,239) codes, yielding a frame size of $255 \cdot 16 \cdot 8 = 32,640$.

In modern FEC, LDPC codes are highly specialized and are not widely interoperable or available in open standards. Private vendors may use proprietary LDPC in FEC despite the high decoding complexity at optical-level throughput (beyond 400G). Overall, LDPC are used in optical communications at the highest competitive level. For instance, Nokia's SDFEC-G1 and SDFEC-G2 proprietary schemes use LDPC codes that attain an extremely high 11.7 dB of coding gain [5].

6. Concluding Remarks

This report surveyed some major landmarks in FEC for optical communications by reviewing literature, industrial standards, and throughput advancements since the 1990s. Despite being a major factor in optical transport networks, FEC is one of many that enable the continued growth of data rates today. Modulation is another factor that significantly affects the performance of an OTN, encoding the channel-coded data into light for physical transmission over the optical channel. Multiplexing (and demultiplexing) enable the "packing" of several independent data streams through one channel by putting each stream on a different wavelength or frequency, making interference impossible. Table 1 summarizes the codes examined in the report, from the original 1996 RS code, to the 2004 super-FEC concatenated code, to the modern era of LDPC, o-FEC, and c-FEC.

	RS(255,239)	super-FEC	LDPC	c-FEC	o-FEC
Gain (dB)	6.2	8.5	11.7	11.2	11.1
Overhead (%)	6.7	6.6	25.0	14.8	15.3

TABLE 1. Comparing FEC Codes at 10^{-15} BER

There is no single best FEC code for optical communication. Selecting an FEC scheme to use today depends on many factors related to the application, for example: travel distance, latency requirement, post-FEC BER requirement, and channel conditions. We have seen that o-FEC is better suited to long-haul bulk optical transport over several hundred to several thousand kilometers, while c-FEC is better suited (and specified for) low-latency applications traveling less than 120 kilometers. LDPC FEC schemes remain an advanced and costly implementation but offer the best achievable coding gain.

This survey demonstrated that as FEC pushes practical implementation toward Shannon limits, elementary and historic codes remain central to all practical schemes. Codes of Reed, Solomon, Hamming, and Gallager are still highly relevant in high-speed optical communication, and will likely continue to be. When paired with sophisticated frameworks like staircasing and zippering, and with methods like iterative Turbo decoding, an optimal balance between theoretical performance and practical applicability is attained.

References

[1] S. Benedetto and G. Bosco. "Channel Coding for Optical Communications". In: *Optical Communication Theory and Techniques*, Springer, Boston, MA, 2005. Chap. 8.

[2] C. Berrou, A. Glavieux, and P. Thitimajshima. "Near Shannon limit error-correcting coding and decoding: Turbo-codes (1)". In: *Proceedings of ICC '93 - IEEE International Conference on Communications*. Vol. 2. 1993, pp. 1064–1070. DOI: 10.1109/ICC.1993.397441.

[3] D. Chase. "A Class of Algorithms for Decoding Block Codes with Channel Measurement Information". In: *IEEE Transactions on Information Theory* 18.1 (Jan. 1972), pp. 170–182.

[4] I. Djordjevic, W. Ryan, and B. Vasic. "Channel Coding for Optical Channels". In: *Coding for Optical Channels*. Boston, MA: Springer US, 2010, pp. 123–178. ISBN: 978-1-4419-5569-2. DOI: 10.1007/978-1-4419-5569-2_4.

[5] R. Eisenach. *What the FEC?* Nokia. Oct. 28, 2022. URL: https://www.nokia.com/blog/what-the-fec/.

[6] S. Forchhammer, J.D. Andersen, and K.J. Larsen. "Enhancing oFEC using MAP Decoding of BCH Component Codes". In: *2025 13th International Symposium on Topics in Coding (ISTC)*. 2025, pp. 1–5. DOI: 10.1109/ISTC65386.2025.11154622.

[7] Optical Interworking Forum. *Implementation Agreement 400ZR: OIF-400ZR-01.0*. Tech. rep. Open ROADM, Mar. 2020.

[8] R. Gallager. "Low-Density Parity-Check Codes". In: *IEEE Transactions on Information Theory* 8.1 (1962), pp. 21–28.

[9] Open ROADM. *Open ROADM MSA 5.0 W-Port Digital Specification (100G-400G)*. Tech. rep. Open ROADM, July 2021.

[10] R. Roth. *Introduction to Coding Theory*. Cambridge University Press, 2006.

[11] International Telecommunications Union - Telecommunications Standards Sector. *ITU-T Recommendation G.709: Interfaces for the optical transport network (OTN)*. Tech. rep. G.975.1. International Telecommunication Union, Telecommunication Standardization Sector (ITU-T), Feb. 2004.

[12] International Telecommunications Union - Telecommunications Standards Sector. *ITU-T Recommendation G.975: Forward error correction for submarine systems*. Tech. rep. G.975. Series G: Transmission systems and media; Digital sections and digital line system; Optical fibre submarine cable systems. International Telecommunication Union, Telecommunication Standardization Sector (ITU-T), Nov. 1996.

[13] International Telecommunications Union - Telecommunications Standards Sector. *ITU-T Recommendation G.975.1: Forward error correction for high bit-rate DWDM submarine systems*. Tech. rep. G.975.1. International Telecommunication Union, Telecommunication Standardization Sector (ITU-T), Feb. 2004.

[14] B. Smith et al. "Staircase Codes: FEC for 100 Gb/s OTN". In: *Journal of Lightwave Technology* 30.1 (2012), pp. 110–117. DOI: 10.1109/JLT.2011.2175479.

[15] A. Y. Sukmadji. "Zipper codes: High-rate spatially-coupled codes with algebraic component codes". MA thesis. Canada: University of Toronto, 2020.

[16] D. Williams. *Turbo Product Code FEC Contribution*. Tech. rep. 802.16.1. IEEE, June 2000.